

claude-windows / 6a82308f-69d6-43a8-9ea4-d8eaf497253d

Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-inde  
xer\6a82308f-69d6-43a8-9ea4-d8eaf497253d\subagents\workflows\wf_3d536535-241\agent-  
a31a864cf541b33fd.jsonl`  
- SHA-256: `0877a1f685fb5ca6f91d9c17a1f15f472d3a1e91c4fa536ebd2d53a5f08b3441`  
- Source modified: `2026-06-16T03:23:54+00:00`  
- Imported at: `2026-07-05T16:48:27+00:00`  
- project: `wf_3d536535-241`  
- session_id: `6a82308f-69d6-43a8-9ea4-d8eaf497253d`
```

Transcript

1. user (2026-06-16T03:21:40.623Z)

Adversarial Claim Verifier (voter 1/3)

Be SKEPTICAL. Try to REFUTE this claim. ?2/3 refutations kill it.

Research question

Research question: ****What concrete, recent (2022?2025) techniques and evaluation practices should a local, commercially-licensed badminton rally-detector adopt to close the quality gap with a strong cloud VLM (Gemini), given a tiny labeled set (~6 ? ~16 golden videos)?**** Produce an adversarially-verified, cited report organized around the four dimensions below. Prefer specific named methods, papers (arXiv/venue), and reported numbers over generic advice; flag anything you cannot verify.

CONTEXT (so you target the real gap, not generic ML):

- System: a local pipeline detects the shuttle per-frame (WASB / TrackNet, MIT-licensed) ? a heuristic turns the trajectory into candidate rally windows ? today an optional Gemini "handoff" confirms rallies. We just measured a fully-local (no-AI) run vs Gemini on 3 matches: local emits far FEWER but MUCH LONGER windows (mean 65s vs 6.1s; one window spanned a whole 174s clip) ? it does NOT miss activity, it FAILS TO SEGMENT. Root cause (code-confirmed): a frame is "active" iff the shuttle is visible AND moving above a small per-second-normalized velocity threshold; active frames within a 2 s gap are merged; there is NO max-window cap and NO inactivity/boundary split. So "shuttle is moving" is conflated with "rally in progress," and dead-time + serves + knock-ups bridge rallies into mega-windows. Shuttle tracking itself is excellent (96?99% per-frame visibility).

- HARD CONSTRAINTS: weights/data/code must be MIT/Apache-2.0/BSD only (no viral/NC/custom licenses, reject Gemma-3/Llama-4-MAU-cap); we may NEVER train owned weights on paid-LLM (Gemini/OpenAI/Anthropic) outputs (terms/copyright), though paid LLMs may be used at inference; minors excluded from any training pool; runs on a single local GPU.

- WHAT WE ALREADY HAVE (do NOT re-survey the basics ? go deeper/newer): the architecture skeleton is known to be standard Temporal Action Localization/Segmentation (proposal+scorer) + late/score-level fusion + stacked generalization; in-domain prior art (MonoTrack, ShuttleSet, See et al. 2024/25 non-broadcast badminton rally start/end, Ghosh et al. point-segmentation, tennis/TT play-break state machines) is already cited. We have a working eval harness: leave-one-video-out (grouped by match), temporal-IoU F1 @0.5, F1@{0.1,0.25,0.5}, mAP@tIoU, segment-count-ratio, bootstrap 95% CIs, paired exact sign-test, Platt/isotonic calibration + Brier/ECE, a tiny numpy LogisticRegression scorer over per-window feature columns, an experiment leaderboard, and an "eval?serve" guard (a cue measured +0.074 F1 on permissive proposal windowing but ?0.008 on the coarser served windowing ? reverted). Available but default-OFF per-window signals: net-crossing count (rally gate), 5 person/court features, 3 audio-hit features.

THE FOUR DIMENSIONS:

1) WINDOWING & BOUNDARIES (Temporal Action Localization/Segmentation): beyond proposal+score ? what are the best ***recent*** methods for (a) turning a dense per-frame signal into well-segmented

actions, (b) explicitly controlling OVER-segmentation/merging, (c) boundary detection/regression and splitting merged segments? Cover e.g. MS-TCN/MS-TCN++, ASRF (action-boundary regression), BMN/BSN boundary-matching, ActionFormer/TriDet/TemporalMaxer (actionness + boundary heads), smoothing/boundary losses, and the standard metrics for over-segmentation (segmental F1@k, edit/Levenshtein score, mAP@tIoU). Which transfer to a 1-D shuttle-trajectory + auxiliary-cue setting on tiny data?

2) SHUTTLE-TRAJECTORY ? RALLY SIGNAL PROCESSING (racket-sports specific): how do published systems convert ball/shuttle tracking into rally/point segments and distinguish "in-play" from dead-time? Cover rally-state machines, serve/let detection, net-crossing & two-sided-occupancy cues, hit/contact (audio + kinematic) detection, and fps-invariant trajectory features. What features best separate "rally" from "shuttle merely moving"?

3) SMALL-DATA STRATEGIES: with ~6?16 labeled videos, what gives the most quality per label ? active learning (which videos/segments to label next; uncertainty/diversity/core-set), semi-/weakly-/point-supervised TAL (e.g. SF-Net, timestamp supervision), self-training/pseudo-labeling, and data-efficient TAL? Separately: the legally-clean ways to exploit a strong teacher (Gemini) at INFERENCE for weak/pseudo-labels or as an evaluation oracle WITHOUT its outputs becoming training data for owned weights ? what does the literature say about teacher-student / distillation boundaries and label-noise handling, and what governance distinctions matter?

4) EVALUATION METHODOLOGY & RIGOR at small n: trustworthy measurement with ~6?16 videos ? bootstrap/permutation tests, paired sign-tests, leave-one-group-out vs RL00CV bias, calibration (reliability, ECE), the over-segmentation-blindness of plain tIoU, shadow/A-B evaluation, regression gates, and concrete pitfalls (label noise, annotator agreement/IAA, multiple-comparison/over-fitting traps, eval-vs-serve drift). What metrics + protocols should a small-data sports-segmentation leaderboard standardize on?

Deliverable: a structured, cited report with, per dimension, the 3?5 highest-leverage techniques that APPLY to our constrained setting (small data, local, license-clean, 1-D trajectory + cues + learned scorer), each with: what it is, why it fits our over-merge/small-data problem, the expected difficulty/payoff, and the citation. Call out which ideas are low-effort high-confidence wins vs research bets. Be explicit about anything unverified or license-incompatible.

Claim under review

"TriDet's Trident-head localizes action boundaries by estimating a relative probability distribution over a set of local 'bins' around the boundary (computing the offset as an expected value over neighboring instants), rather than regressing a boundary offset from a single instant or from a segment's global feature ? directly addressing imprecise/ambiguous boundary localization."

****Source:**** https://openaccess.thecvf.com/content/CVPR2023/papers/Shi_TriDet_Temporal_Action_Detection_With_Relative_Boundary_Modeling_CVPR_2023_paper.pdf (primary)

****Supporting quote:**** "instead of directly predicting the boundary offsets based on the center point feature, the proposed Trident-head models the action boundary via an estimated relative probability distribution of the boundary (see Fig. 1). The boundary offset is then computed based on the expected values of neighboring locations (i.e. bins)."

Checklist

1. Is the claim actually supported by the quote, or is it an overreach/misread?
2. WebSearch for contradicting evidence ? does any credible source dispute or heavily qualify this?
3. Is the source quality sufficient for the claim's strength? (extraordinary claims need primary sources)
4. Is the claim outdated? (check dates ? old claims about fast-moving fields are suspect)
5. Is this a marketing claim / press release / cherry-picked benchmark / forum speculation?

****refuted=true**** if: unsupported by quote / contradicted / low-quality source for strong claim / outdated / marketing fluff.

****refuted=false**** ONLY if: claim is well-supported, current, and source quality matches claim strength.

Default to refuted=true if uncertain.

Structured output only. Evidence MUST be specific.

2. assistant (2026-06-16T03:21:44.242Z)

I'll verify this claim about TriDet's Trident-head boundary modeling. Let me fetch the primary source.

3. assistant (2026-06-16T03:21:54.163Z)

I'll fetch the primary source to verify this specific claim about TriDet's Trident-head mechanism.

4. user (2026-06-16T03:22:02.851Z)

The server returned HTTP 403 Forbidden.

The response body was not retrieved. If this URL requires authentication, use an authenticated tool (e.g. `gh` for GitHub, or an MCP-provided fetch tool) instead of WebFetch.

5. assistant (2026-06-16T03:22:05.387Z)

The CVF PDF is blocking direct fetch. Let me try the arXiv version and search for corroborating sources.

6. user (2026-06-16T03:22:11.663Z)

TriDet Abstract and Trident-head Mechanism

Abstract

The paper presents TriDet, a one-stage temporal action detection framework. The authors state: "we propose a novel Trident-head to model the action boundary via an estimated relative probability distribution around the boundary."

Analysis of Your Specific Questions

Based on the provided content, I can only partially address your questions:

(1) Relative probability distribution over local bins:

Yes, this is confirmed. The abstract explicitly describes the mechanism as modeling "the action boundary via an estimated relative probability distribution around the boundary."

(2) & (3) Expected value computation and contrast with direct regression:

Unfortunately, the provided webpage content does not include sufficient technical detail to confirm whether:

- The boundary offset is computed as an expected value over neighboring bins
- The paper explicitly contrasts this approach with direct offset regression from a single point

The abstract does not elaborate on these implementation specifics. To answer these questions fully, you would need to access the complete paper PDF, which is referenced but not included in the webpage content provided.

7. user (2026-06-16T03:22:15.307Z)

Web search results for query: "TriDet Trident-head relative boundary modeling probability distribution bins expected value offset CVPR 2023"

Links: [{"title": "TriDet - CVPR 2023 Open Access Repository", "url": "https://openaccess.thecvf.com/content/CVPR2023/html/Shi_TriDet_Temporal_Action_Detection_With_Relative_Boundary_Modeling_CVPR_2023_paper.html"}, {"title": "TriDet: Temporal Action Detection", "url": "https://www.emergentmind.com/papers/2303.07347"}, {"title": "TriDet: Temporal Action Detection with Relative Boundary Modeling | IEEE Conference Publication | IEEE Xplore", "url": "https://ieeexplore.ieee.org/document/10203543/"}, {"title": "[2303.07347] TriDet: Temporal Action Detection with Relative Boundary Modeling", "url": "https://arxiv.org/abs/2303.07347"}, {"title": "CVPR Poster TriDet: Temporal"}]

Action Detection With Relative Boundary Modeling", "url": "https://cvpr.thecvf.com/virtual/2023/poster/21956"}, {"title": "TriDet: Temporal Action Detection with Relative Boundary Modeling - ADS", "url": "https://ui.adsabs.harvard.edu/abs/2023arXiv230307347S/abstract"}, {"title": "TriDet: Temporal Action Detection with Relative Boundary Modeling Dingfeng Shi?", "url": "https://openaccess.thecvf.com/content/CVPR2023/papers/Shi_TriDet_Temporal_Action_Detection_With_Relative_Boundary_Modeling_CVPR_2023_paper.pdf"}, {"title": "TriDet: Temporal Action Detection with Relative Boundary Modeling", "url": "https://arxiv.org/pdf/2303.07347"}, {"title": "[Quick Review] TriDet: Temporal Action Detection with Relative Boundary Modeling", "url": "https://liner.com/review/tridet-temporal-action-detection-with-relative-boundary-modeling"}, {"title": "GitHub - dingfengshi/TriDet: [CVPR2023] Code for the paper, TriDet: Temporal Action Detection with Relative Boundary Modeling . GitHub", "url": "https://github.com/dingfengshi/TriDet"}]

Based on the search results, I found detailed information about TriDet and its Trident-head approach for CVPR 2023:

TriDet: Temporal Action Detection with Relative Boundary Modeling

TriDet is a one-stage framework for temporal action detection that proposes a novel Trident-head to model the action boundary via an estimated relative probability distribution around the boundary.

Key Technical Details

The Trident-head consists of three components: a start head, an end head, and a center-offset head, which are designed to locate the start boundary, end boundary, and the temporal center of the action, respectively.

The predicted boundary response in neighboring bin set is summed element-wise with the center offset corresponding to each instant, which is further estimated as the relative boundary distribution. The offset is then computed based on the expected value of the bin.

Performance

TriDet achieves state-of-the-art performance on three challenging benchmarks: THUMOS14, HACS and EPIC-KITCHEN 100, with lower computational costs. For example, TriDet hits an average mAP of 69.3% on THUMOS14, outperforming the previous best by 2.5%, but with only 74.6% of its latency.

Additional Innovation

In the feature pyramid of TriDet, they propose a Scalable-Granularity Perception (SGP) layer to aggregate information across different temporal granularities, which is much more efficient than recent transformer-based feature pyramids.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

8. assistant (2026-06-16T03:22:17.894Z)

Let me get the verbatim arXiv PDF text to confirm the exact mechanism, especially the "expected value over bins" and the contrast with single-instant/center-point regression.

9. user (2026-06-16T03:22:26.647Z)

I appreciate your detailed request, but I need to be transparent: the content provided appears to be raw PDF object streams (compressed binary data), not readable text. The material shows PDF structure objects, image data, and encoded font information, but doesn't contain the actual textual content in a form I can search and quote from.

To properly locate and quote the sentences describing the Trident-head mechanism, I would need:

1. **Actual readable text** from the document (not PDF binary encoding)

2. ****Direct access to the Method section**** describing the three heads (start, end, center-offset)

3. ****The specific passages**** near Figure 1 explaining boundary offset computation

****What I can observe:**** The PDF references figures like "intro.pdf," "cosine.pdf," and "framework.pdf," which suggests this is an academic paper with technical diagrams, but the text layer isn't accessible in the format provided.

****Recommendation:**** Please share the document either as:

- Plain text extracted from the PDF
- A readable markdown/HTML version
- The actual PDF file interpreted through a text-extraction tool

Once you provide readable text, I'll be happy to locate those exact sentences about the Trident-head mechanism while respecting the 125-character quote limit.

[Binary content (application/pdf, 1.4MB) also saved to C:\Users\avidu\claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\6a82308f-69d6-43a8-9ea4-d8eaf497253d\tool-results\webfetch-1781579910495-xf7r2b.pdf]

10. assistant (2026-06-16T03:22:28.392Z)

The PDF was saved locally. Let me read it directly to extract verbatim text.

11. user (2026-06-16T03:22:29.474Z)

pdftoppm failed: Command 'pdftoppm' not found or is in an unsafe location (current directory)

12. assistant (2026-06-16T03:22:31.516Z)

Let me extract the PDF text using Python.

13. user (2026-06-16T03:22:35.130Z)

===PAGE 1===

TriDet: Temporal Action Detection with Relative Boundary Modeling

Dingfeng Shi?

VRLab, Beihang University, China

shidingfeng@buaa.edu.cn

Yujie Zhong

Meituan Inc.

jaszhong@hotmail.com

Qiong Cao?

JD Explore Academy

mathqiong2012@gmail.com

Lin Ma

Meituan Inc.

forest.linma@gmail.com

Jia Li?

VRLab, Beihang University, China

jiali@buaa.edu.cn

Dacheng Tao

JD Explore Academy

dacheng.tao@gmail.com

Abstract

In this paper, we present a one-stage framework TriDet for temporal action detection. Existing methods often suffer from imprecise boundary predictions due to the ambiguous action boundaries in videos. To alleviate this problem, we propose a novel Trident-head to model the action boundary via an estimated relative probability distribution around the boundary. In the feature pyramid of TriDet, we propose an efficient Scalable-Granularity Perception (SGP) layer to mitigate the rank loss problem of self-attention that

takes place in the video features and aggregate information across different temporal granularities. Benefiting from the Trident-head and the SGP-based feature pyramid, TriDet achieves state-of-the-art performance on three challenging benchmarks: THUMOS14, HACS and EPIC-KITCHEN 100, with lower computational costs, compared to previous methods. For example, TriDet hits an average mAP of 69.3% on THUMOS14, outperforming the previous best by 2.5%, but with only 74.6% of its latency. The code is released to <https://github.com/dingfengshi/TriDet>.

1. Introduction

Temporal action detection (TAD) aims to detect all start and end instants and corresponding action categories from an untrimmed video, which has received widespread attention. TAD has been significantly improved with the help of the deep learning. However, TAD remains to be a very challenging task due to some unresolved problems.

A critical problem in TAD is that action boundaries are usually not obvious. Unlike the situation in object detection where there are usually clear boundaries between the

*: This work is done during an internship at JD Explore Academy.

?: Corresponding authors.

Segment-level

Instant-level

Ours

Global

FeatureStart End

Start End

Other

Feature

Refine

Temporal

Feature

Temporal

Feature

Start

Bin Set Bin Set

EndTemporal

Feature

Figure 1. Illustration of different boundary modeling. Segment-level: these methods locate the boundaries based on the global feature of a predicted temporal segment. Instant-level: they directly regress the boundaries based on a single instant, potentially with some other features. Ours: the action boundaries are modeled via an estimated relative probability distribution of the boundary. objects and the background, the action boundaries in videos can be fuzzy. A concrete manifestation of this is that the instants (i.e. temporal locations in the video feature sequence) around the boundary have relatively higher predicted response value from the classifier.

Some previous works attempt to locate the boundaries based on the global feature of a predicted temporal segment [22,23,30,48,53], which may ignore detailed information at each instant. As another line of work, they directly regress the boundaries based on a single instant [33,49], potentially with some other features [21, 34, 51], which do not consider the relation between adjacent instants (e.g. the rel-

arXiv:2303.07347v2 [cs.CV] 16 Mar 2023

===PAGE 2===

0.60

0.70

0.80

0.90

1.00

0 1 2 3 4 5 6

Cosine Similarity

Layers

SA

SGP

BackB.

Figure 2. Within the HACS dataset and SlowFast backbone, we statistic the average cosine similarity between features at each instant and the video-level average feature for self-attention and SGP, respectively. We observe that the SA exhibits high similarity, indicating poor discriminability (i.e. rank loss problem). In contrast, SGP resolves the issue and exhibits stronger discriminability. active probability) around the boundary. How to effectively utilize boundary information remains an open question.

To facilitate localization learning, we posit that the relative response intensity of temporal features in a video can mitigate the impact of video feature complexity and increase localization accuracy. Motivated by this, we propose a one-stage action detector with a novel detection head named Trident-head tailored for action boundary localization. Specifically, instead of directly predicting the boundary offsets based on the center point feature, the proposed Trident-head models the action boundary via an estimated relative probability distribution of the boundary (see Fig. 1). The boundary offset is then computed based on the expected values of neighboring locations (i.e. bins).

Apart from the Trident-head, in this work, the proposed action detector consists of a backbone network and a feature pyramid. Recent TAD methods [10, 42, 49] adopt the transformer-based feature pyramid and show promising performance. However, the video features of the video backbone tend to exhibit high similarities between snippets, which is further deteriorated by SA, leading to the rank loss problem [13] (see Fig. 2). Additionally, SA also incurs significant computational overhead.

Fortunately, we discover that the success of the previous transformer-based layers (in TAD) primarily relies on their macro-architecture, namely, how the normalization layer and feed-forward network (FFN) are connected, rather than the self-attention mechanism. We therefore propose an efficient convolutional-based layer, termed Scalable-Granularity Perception (SGP) layer, to alleviate the two abovementioned problems of self-attention. SGP comprises two primary branches, which serve to increase the discrimination of features in each instant and capture temporal information with different scales of receptive fields.

The resultant action detector is termed TriDet. Extensive experiments demonstrate that TriDet surpasses all the previous detectors and achieves state-of-the-art performance across three challenging benchmarks: THUMOS14, HACS and EPIC-KITCHEN 100.

2. Related Work

Temporal action detection. Temporal action detection (TAD) involves localizing and classifying all actions from an untrimmed video. The existing methods can be roughly divided into two categories, namely, two-stage methods and one-stage methods. The two-stage methods [34, 37, 45, 48, 55] split the detection process into two stages: proposal generation and proposal classification. Most of the previous works [9, 14, 20, 22, 23, 27] put emphasis on the proposal generation phrase. Concretely, some works [9,22,23] predict the probability of the action boundary and densely match the start and end instants according to the prediction score. Anchor-based methods [14,20] classify actions from specific anchor windows. However, two-stage methods suffer from a high complexity problem and can not be trained

in an end-to-end manner. The one-stage methods do the localization and classification with a single network. Some previous works [21, 46, 47] build this hierarchical architecture with the convolutional network (CNN). However, there is still a performance gap between the CNN-based and the latest TAD methods.

Object detection. Object detection is a twin task of TAD. General Focal Loss [19] transforms bounding box regression from learning Dirac delta distribution to a general distribution function. Some methods [11, 16, 29] use Depth-wise Convolution to model network structure and some branched designs [17, 38] show high generalization ability. They are enlightening for the architecture design of TAD.

Transformer-based methods. Inspired by the great success of the Transformer in the field of machine translation and object detection, some recent works [10, 26, 28, 36, 39, 49] adopt the attention mechanism in TAD task, which help improve the detection performance. For example, some works [28, 36, 39] detect the action with the DETR-like Transformer-based decoder [7], which models action instances as a set of learnable. Other works [10, 49] extract a video representation with a Transformer-based encoder. However, most of these methods are based on the local behavior. Namely, they conduct attention operation only in a local window, which introduces an inductive bias similar to CNN but with a larger computational complexity and additional limitations (e.g. The length of the sequence needs to be pre-padded to an integer multiple of the window size.).

3. Method

Problem definition. We first give a formal definition for TAD task. Specifically, given a set of untrimmed videos

$D = \{V_i\}_n$

$i=1$, we have a set of RGB (and optical flow)

===PAGE 3===

Detection Head

Classification Head

1

1/2

1/4

1/8

1/16

1/32

1/64

SGP Feature Pyramid

Start

Boundary

Center

Offset

End

Boundary

Trident-head

SGP Layer

SGP Layer

SGP Layer

SGP Layer

SGP Layer

SGP Layer

SGP Layer

Backbone

Pyramid Feature

Start

Boundary

End

Boundary

Figure 3. Illustration of TriDet. We build the pyramid features with Scalable-Granularity

Perception (SGP) layer. The corresponding features in each level are fed into a shared-weight detection head to obtain the detection result, which consists of a classification head and a Trident-head. The Trident-head estimates the boundary offset based on a relative distribution predicted by three branches: Start Boundary, End Boundary and Center Offset.

temporal visual features $X_i = \{x_t\}_T$

$t=1$ from each video V_i ,

where T corresponds to the number of instants, and K_i segment labels $Y_i = \{s_k, e_k, c_k\}_{K_i}$

$k=1$ with the action segment

start instant s_k , the end instant e_k and the corresponding action category c_k . TAD aims at detecting all segments Y_i based on the input feature X_i .

3.1. Method Overview

Our goal is to build a simple and efficient one-stage temporal action detector. As shown in Fig. 3, the overall architecture of TriDet consists of three main parts: a video feature backbone, a SGP feature pyramid, and a boundary-oriented Trident-head. First, the video features are extracted using a pretrained action classification network (e.g. I3D [8] or SlowFast [15]). Following that, a SGP feature pyramid is built to tackle actions with various temporal lengths, similar to some recent TAD works [10,21,49]. Namely, the temporal features are iteratively downsampled and each scale level is processed with a proposed Scalable-Granularity Perception (SGP) layer (Section 3.2) to enhance the interaction between features with different temporal scopes. Lastly, action instances are detected by a designed boundary-oriented Trident-head (Section 3.3). We elaborate on the proposed modules in the following.

3.2. Feature Pyramid with SGP Layer

The feature pyramid is obtained by first downsampling the output features of the video backbone network several times via max-pooling (with a stride of 2). The features at each pyramid level are then processed using transformer-like layers (e.g. ActionFormer [49]).

Current Transformer-based methods for TAD tasks primarily rely on the macro-architecture of the Transformer (See supplementary material for details), rather than the self-attention mechanisms. Specifically, SA mainly encounters two issues: the rank loss problem across the temporal dimension and its high computational overhead.

Limitation 1: the rank loss problem. The rank loss problem arises because the probability matrix in self-attention (i.e. $\text{softmax}(QK^T)$) is non-negative and the sum of each row is 1, indicating the outputs of SA are convex combination for the value feature V . Considering that pure Layer Normalization [3] projects feature onto the unit hyper-sphere in high-dimensional space, we analyze the degree of their distinguishability by studying the maximum angle between features within the instant features. We demonstrate that the maximum angle between features after the convex combination is less than or equal to that of the input features, resulting in increasing similarity between features (as outlined in the supplementary material), which can be detrimental to TAD.

Limitation 2: high computational complexity. In addition, the dense pair-wise calculation (between instant features) in self-attention brings a high computational overhead and therefore decreases the inference speed.

The SGP layer. Based on the above discovery, we propose a Scalable-Granularity Perception (SGP) layer to effectively capture the action information and suppress rank loss. The major difference between the Transformer layer

and SGP layer is the replacement of the self-attention module with the fully-convolutional module SGP. The successive Layer Normalization [3] (LN) is changed to Group Normalization [43] (GN).

As shown in Fig. 4, SGP contains two main branches: an instant-level branch and a window-level branch. In the

===PAGE 4===

?SGP GNLN FFNx ?

x

AvgPool

FC + ReLU

FC

14. assistant (2026-06-16T03:23:21.741Z)

I now have the verbatim primary-source text. Let me get the detailed Method section (3.3) describing the bins and expected-value computation to fully confirm.

15. user (2026-06-16T03:23:27.403Z)

?SGP GNLN FFNx ?

x

AvgPool

FC + ReLU

FC

Conv_w

Conv_{kw}

Conv_w

?

?

??

Instant-level

Window-level

Shortcut

?Self-

attention LNLN FFNx ?

Transformer Layer

SGP Layer

??

??

??????

????????

Figure 4. Illustration of the structure of SGP layer. We replace the self-attention and the second Layer Normalization (LN) with SGP and Group Normalization (GN), respectively.

instant-level branch, we aim to increase the feature discriminability between action and non-action instant by enlarging their feature distance with the video-level average feature.

The window-level branch is designed to introduce the semantic content from a wider receptive field with a branch ? to help dynamically focus on the features of which scale.

Mathematically, the SGP can be written as:

$f_{SGP} = ?(x)F C(x) + ?(x)(Conv w(x) + Conv kw(x)) + x,$

(1)

where F and Conv w denotes fully-connected layer and the 1-D depth-wise convolution layer [11] over temporal dimension with window size w. As a signature design of SGP, k is a scalable factor aiming at capturing a larger granularity of temporal information. The video-level average feature

$?(x)$ and branch $?(x)$ are given as

$?(x) = ReLU(F C(AvgPool(x)))$, (2)

$?(x) = Conv w(x)$, (3)

where AvgPool(x) is the average pooling for all features over the temporal dimension. Here, both $?(x)$ and $?(x)$ perform the element-wise multiplication with the main-

stream feature.

The resultant SGP-based feature pyramid can achieve better performance than the transformer-based feature pyramid while being much more efficient.

3.3. Trident-head with Relative Boundary Modeling

Intrinsic property of action boundaries. Regarding the detection head, some existing methods directly regress the

?

Softmax

?????? = $E_{t \sim \text{??????}} = ??$. ??

Start Prediction

Center Offset

Video Feature

Reference Times

Relative Probability of Start Boundary

offset

Bin Set

Action Start Ground-truth Action T

??

??????

012345

????

??,??

t-B ?? ????

??

Figure 5. The boundary localization mechanism of Trident-head.

We predict the boundary response and the center offset for each instant. At the instant t , the predicted boundary response in neighboring bin set is summed element-wise with the center offset corresponding to the instant t , which is further estimated as the relative boundary distribution. Finally, the offset is computed based on the expected value of the bin.

temporal length [49] of the action at each instant of the feature and refine with the boundary feature [21, 34], or [22, 23, 48] simply predict an actionness score (indicating the probability of being an action). These simple strategies suffer from a problem in practice: imprecise boundary predictions, due to the intrinsic property of actions in videos.

Namely, the boundaries of actions are usually not obvious, unlike the boundaries of objects in object detection. Intuitively, a more statistical boundary localization method can reduce uncertainty and facilitate more precise boundaries.

Trident-head. In this work, we propose a boundary-oriented Trident-head to precisely locate the action boundaries based on the relative boundary modeling, i.e. considering the relation of features in a certain period and obtaining the relative probability of being a boundary for each instant in that period. The Trident-head consists of three components: a start head, an end head, and a center-offset head, which are designed to locate the start boundary, end boundary, and the temporal center of the action, respectively. The Trident-head can be trained end-to-end with the detector.

Concretely, as shown in Fig. 5, given a sequence of features $F \in \mathbb{R}^{T \times D}$ output from the feature pyramid, we first obtain three feature sequences from the three branches (namely, $F_s \in \mathbb{R}^{T \times B}$, $F_e \in \mathbb{R}^{T \times B}$ and $F_c \in \mathbb{R}^{T \times 2 \times (B+1)}$),

where B is the number of bins for boundary prediction, F_s and F_e characterize the response value for each instant as the starting or ending point of an action, respectively. In addition, the center-offset head aims at estimating two conditional distributions $P(bst|t)$ and $P(bet|t)$. They represent the probability that each instant (in its set of bins) serves as a boundary when the instant t is the midpoint of an action.

Then, we model the boundary distance by combining the outputs of the boundary head and center-offset head:

```

ePst = Sof tmax(F [(t?B):t]
s + F t,0
c ), (4)
dst = Eb? ePst
[b] ?
BX
b=0
(b ePstb), (5)
where F [(t?B):t]
s ? RB+1, F t,0
c ? RB+1 are the feature
of the left adjacent bin set of instant t and the center off-
sets predicted by instant t only, respectively, and ePst is the
relative probability which represents the probability of each
instant as a start of the action within the bin set. Then, the
distance between the instant t and the start instant of action
instance dst is given by the expectation of the adjacent bin
set. Similarly, the offset distance of the end boundary det
can be obtained by

```

```

ePet = Sof tmax(F [t:(t+B)]
e + F t,1
c ), (6)
det = Eb? ePet
[b] ?

```

```

BX
b=0
(b ePetb) (7)

```

All heads are simply modeled in three layers convolutional networks and share parameters at all feature pyramid levels to reduce the number of parameters. Combination with feature pyramid. We apply the Trident-head in a pre-defined local bin set, which can be further improved by combining it with the feature pyramid. In this setting, features at each level of the feature pyramid simply share the same small number of bins B (e.g. 16) and then the corresponding prediction for each level l can be scaled by 2^{l-1} , which can significantly help to stabilize the training process.

Formally, for an instant in the l -th feature level t_l , TriDet estimates the boundary distance d_l and d_l with

the Trident-head described above, then the segments $a = (s_t, e_t)$ can be decoded by

```

s_t = (t - d_l
s_t) × 2^{l-1}, (8)

```

```

e_t = (t + d_l
e_t) × 2^{l-1}. (9)

```

Comparison with existing methods that have explicit boundary modeling. Many previous methods improve

16. assistant (2026-06-16T03:23:40.898Z)

I now have the complete, verbatim primary-source evidence. Every element of the claim is directly confirmed.

The claim has four sub-assertions, all verified against the CVPR 2023 primary source (arXiv 2303.07347):

1. *****estimates a relative probability distribution over a set of local 'bins' around the boundary***** ? Confirmed verbatim. Abstract: "model the action boundary via an estimated relative probability distribution around the boundary." Method 3.3: "obtaining the relative probability of being a boundary for each instant in that period"; Eq. (4)/(6) define $\tilde{p}_{st} = \text{Softmax}(\dots)$ as "the relative probability ... within the bin set." B is explicitly "the number of bins for boundary prediction."

2. *****"computing the offset as an expected value over neighboring instants/bins"***** ? Confirmed verbatim. Intro p.2: "The boundary offset is then computed based on the expected values of neighboring locations (i.e. bins)." Eq. (5): $\text{dst} = E_{\{b-Pst\}}[b] \cdot (b - Pst_b)$? literally the expectation over the bin set. Fig. 5 caption: "the offset is computed based on the expected value of the bin."

3. *****"rather than regressing a boundary offset from a single instant or from a segment's global feature"***** ? Confirmed verbatim. Intro p.2: "instead of directly predicting the boundary offsets based on the center point feature." Fig. 1 caption explicitly contrasts three modes: Segment-level ("locate the boundaries based on the global feature of a predicted temporal segment"), Instant-level ("directly regress the boundaries based on a single instant"), and "Ours" (relative probability distribution). Method 3.3 also contrasts with methods that "directly regress the temporal length."

4. *****"directly addressing imprecise/ambiguous boundary localization"***** ? Confirmed verbatim. Abstract: "Existing methods often suffer from imprecise boundary predictions due to the ambiguous action boundaries in videos. To alleviate this problem, we propose a novel Trident-head."

The supporting quote in the claim ("instead of directly predicting the boundary offsets based on the center point feature... The boundary offset is then computed based on the expected values of neighboring locations (i.e. bins)") is an exact match to the paper. The claim is well-supported, current (CVPR 2023, within the 2022-2025 window), and sourced from the primary venue paper. No contradicting evidence exists. The only trivially loose wording in the claim is "from a single instant" combined with "segment's global feature" ? the paper's contrast is primarily against "center point feature" (the instant-level regression) and Fig. 1 separately calls out the segment-global-feature variant; the claim accurately captures both. No overreach.

17. user (2026-06-16T03:23:49.413Z)

Structured output provided successfully

18. assistant (2026-06-16T03:23:54.221Z)

Claim verified as accurate ? refuted=false, high confidence. All four sub-assertions match the primary CVPR 2023 source verbatim (extracted from the arXiv PDF at C:\Users\avidu\claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\6a82308f-69d6-43a8-9ea4-d8eaf497253d\tool-results\webfetch-1781579910495-xf7r2b.pdf).